`Decentralized meSch`: Multi-Agent Energy-Aware Scheduling Under Different Communication Graphs

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I. Introduction

II. Problem Formulation

A. Notation

Let $\mathbb{Z}_0 = \{0, 1, 2, ...\}$ and $\mathbb{Z}_+ = \{1, 2, 3, ...\}$. Let $\mathbb{R}$, $\mathbb{R}_{\geq 0}$, $\mathbb{R}_{>0}$ be the set of reals, non-negative reals, and positive reals respectively. Let $\mathcal{S}^n_{++}$ denote set of symmetric positive-definite matrices in $\mathbb{R}^{n\times n}$. Let $\mathcal{N}(\mu, \Sigma)$ denote a normal distribution with mean μ and covariance $\Sigma \in \mathcal{S}^n_{++}$. The $Q \in \mathbb{S}^n_{++}$ norm of a vector $x \in \mathbb{R}^n$ is denoted $\|x\|_Q = \sqrt{x^T Q x}$.

B. System Description

Consider a multi-agent system, in which each robotic system $i \in \mathcal{R} = \{0, 1, \cdots, N-1\}$ comprises the robot and battery discharge dynamics:

$$\dot{\chi}^i = \begin{bmatrix} \dot{x}^i \\ \dot{e}^i \end{bmatrix} = f^i(\chi^i, u^i) = \begin{bmatrix} f^i_r(x^i, u^i) \\ f^i_e(e^i) \end{bmatrix}, \tag{1}$$

where $N = |\mathcal{R}|$ is the cardinality of the set $\mathcal{R}$, $\chi^i = \begin{bmatrix} x^{iT}, & e^i \end{bmatrix}^T \in \mathcal{Z}^i_r \subset \mathbb{R}^{n+1}$ is the i^{th} robotic system state consisting of the robot state $x^i \in \mathcal{X}^i_r \subset \mathbb{R}^n$ and its State-of-Charge (SoC) $e^i \in \mathbb{R}_{\geq 0}$. $u^i \in \mathcal{U}^i_r \subset \mathbb{R}^m$ is the control input, $f^i : \mathcal{Z}^i_r \times \mathcal{U}^i_r \to \mathbb{R}^{n+1}$ defines the continuous-time robotic system dynamics, $f^i_r : \mathcal{X}^i_r \times \mathcal{U}^i_r \to \mathbb{R}^n$ define robot dynamics and $f^i_e : \mathbb{R}_{\geq 0} \to \mathbb{R}$ define worst-case battery discharge dynamics.

C. Communication Graphs

Consider an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with a set of vertices $\mathcal{V} = 1, \cdots, N$ representing robots and edges $\mathcal{E} = \{\cdots, (i,j), \cdots\}$ representing a communication link between robots i and $j \in \mathcal{R}$. The neighboring set of robot $i \in \mathcal{R}$ represents the set of robots sharing a communication link with the robot i which we denote as $\mathcal{N}_i = \{\forall j \in \mathcal{R} | (i,j) \in \mathcal{E}\}$.

Definition 1. *(Complete graph) The communication graph is complete if a direct communication link exists between all robots i.e.* $(i,j) \in \mathcal{E}\ \forall i, j \in \mathcal{R}$.

Definition 2. *(Connected graph) The communication graph is connected if a communication path exists between any pair of robots, possibly involving multiple communication links.*

Definition 3. *(Time-varying Connected graph) The communication graph is a time-varying connected graph* $G_t = (\mathcal{V}, \mathcal{E}_t)$ *if the edges (communication links) between the vertices (robots), can change over time resulting in a time-varying communication path between any pair of robots* $i, j \in \mathcal{R}$.

D. Problem statement

Consider a team of N robots performing a persistent mission. These robots, modeled by (1), require recharging during the mission. We assume a high-level decentralized planner, which generates a high-level trajectory for each robot based on the high-level mission objectives. We pursue the following two objectives for the robots:

1) adhere to the nominal trajectories and deviate from them only when returning for recharging.
2) ensure mutually exclusive use of the static charging station.

Before formulating an optimization problem, we define $\mathcal{T}^i$ as the set of times i^{th} robot returns to the charging station:

$$\mathcal{T}^i = \{t^i_{s,0}, t^i_{s,1}, \cdots, t^i_{s,m}, \cdots\}, \forall i \in \mathcal{R}, \forall m \in \mathbb{Z}_0 \tag{2}$$

where $t^i_{s,m}$ denotes the m^{th} time i^{th} rechargeable robot returns to the charging station. Then, let $\mathcal{T} = \cup_{i\in\mathcal{R}} \mathcal{T}^i$ be the union of return times for all robots. Subsequently, the trajectory optimization problem is formulated as follows:

$$\min_{\chi^i(t), u^i(t), \mathcal{T}} \sum_{i=0}^{N-1} \int_{t_0}^{\infty} \Big[\left\| x^i(\tau) - x^{i,nom}_0(\tau) \right\|^2_{\mathbf{Q}} + \left\| u^i(\tau) \right\|^2_{\mathbf{R}} \Big] d\tau + |t^{i_1}_{s,m_1} - t^{i_2}_{s,m_2}| \quad \forall t^{i_1}_{s,m_1}, t^{i_2}_{s,m_2} \in \mathcal{T} \tag{3a}$$

$$\text{s.t.} \quad \chi^i(t_0) = \chi^i_0 \qquad \forall i \in \mathcal{R} \tag{3b}$$

$$\dot{\chi}^i = f(\chi^i, u^i) \qquad \forall i \in \mathcal{R} \tag{3c}$$

$$e^i(t) \geq e^i_{min} \qquad \forall i \in \mathcal{R} \tag{3d}$$

$$|t^{i_1}_{s,m_1} - t^{i_2}_{s,m_2}| > T_{ch} + T_\delta \quad \forall t^{i_1}_{s,m_1}, t^{i_2}_{s,m_2} \in \mathcal{T} \tag{3e}$$

where $x^{i,nom}_0 = x^i([t_0, \infty]; t_0, x_0)$ refers to the nominal trajectory of the i^{th} robot generated by the high-level decentralized trajectory planner at the initial time t_0 and state x^i_0. T_{ch} is the charging time, and T_δ is the buffer time. The cost functional (3a) has three components. The first and second terms minimize the state and control costs, respectively. The third term aims to minimize the time interval between the start of charging for any two robots. This is to encourage the robots to follow the nominal trajectory as much as possible before returning for recharging. The constraint (3d) specifies the minimum battery SoC constraints for all robots. (3e) sets the lower bound for the time interval between the return of any two robots. This constraint ensures that only one rechargeable robot is present at the charging station at a given time.

Given the high-level trajectory optimization problem (3), we state three problem variants to be solved under different communication graphs.

Problem 1. *Solve the optimization problem* (3) *in a decentralized setting, assuming a* ***complete communication graph*** *for the robots as defined in Definition 1.*

Problem 2. *Solve the optimization problem* (3) *in a decentralized setting, assuming a* ***connected communication graph*** *for the robots as defined in Definition 2.*

Problem 3. *Solve the optimization problem* (3) *in a decentralized setting, assuming a* ***time-varying connected communication graph*** *for the robots as defined in Definition 3.*